

Growing Like Norway

Inequality, Entrepreneurs, and Innovation

Yikai Wang, Yu Zheng

University of Essex, Queen Mary University

June 10, 2024 @ Fabulous Macro, CUHK

Nordic Countries: Stereotypes

- Low inequality
 - income, wealth
- High security
 - education, health, job
- Low incentives
 - for work, entrepreneurial activities and innovation
- Equality against efficiency
 - Can't We All Be More Like Scandinavians? Acemoglu, Robinson and Verdier (2013)

Income Inequality

Income Gini, after tax

- US, Hong Kong: ~ 0.4
- Italy, Switzerland: ~ 0.33
- Norway, Denmark:

Income Inequality

Income Gini, after tax

- US, Hong Kong: ~ 0.4
- Italy, Switzerland: ~ 0.33
- Norway, Denmark: ~ 0.28

Wealth Inequality

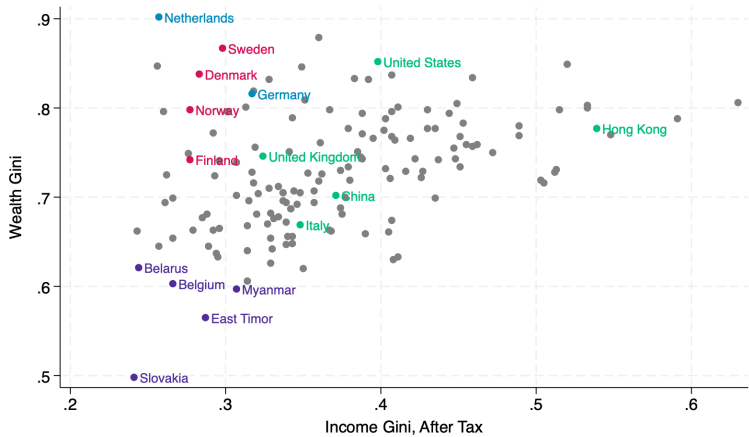
Wealth Gini

- US, Hong Kong, Switzerland: ~ 0.8
- Italy, France : ~ 0.7
- Norway, Denmark:

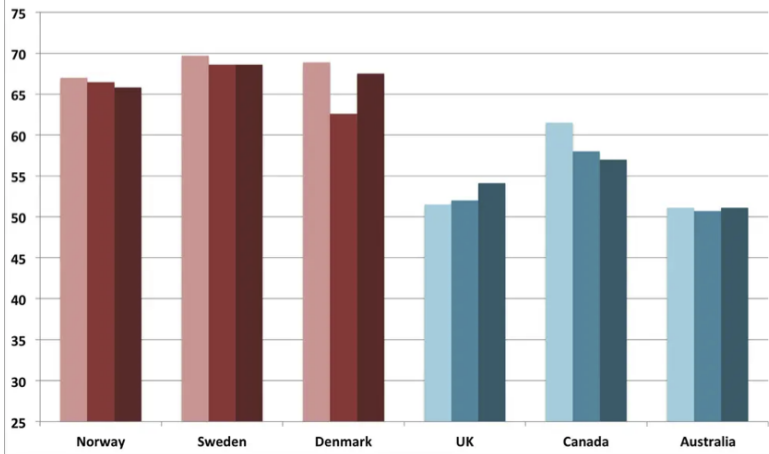
Wealth Inequality

Wealth Gini

- US, Hong Kong, Switzerland: ~ 0.8
- Italy, France : ~ 0.7
- Norway, Denmark: $\sim 0.8!$

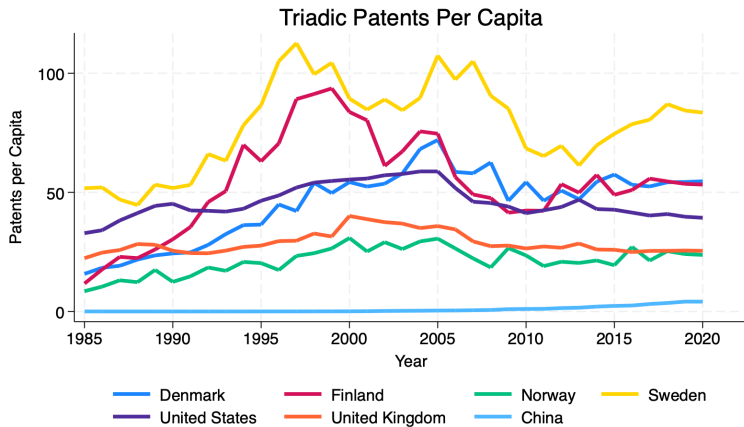


The wealth share of the top decile by country, in 2000, 2007, 2014



Entrepreneurial Activities and Innovation

- Siri
- Safety belt (Volvo)
- Ikea, Nokia
- Minecraft, Angry Birds, Supercell



Research Questions

- Why are wealth inequality high, given the low income inequality?
- Why are entrepreneurial activities and innovation high, in an equal society (lack of incentives)?
- Is equality always against efficiency?
 - Bernie Sanders' Danish Dream vs American Dream: health/education

Main Story of Nordic Model: Inequality

- Low income inequality
 - between low and high skilled workers, not entrepreneurs
- High income of low skilled workers
 - safety net for entrepreneurs, in case fail
- Low wage of high skilled workers
 - high return for entrepreneurs to use high skilled workers
- More entrepreneurial activities, innovation and growth
- Large wealth inequality

Other Parts of Nordic Model: Safety Net and Tax

- Low return of education/high skill?
 - free education/health care
 - low expenditure risks for entrepreneurs and high skilled labor
- Low tax for the (wealth) rich!
 - low capital income tax (22%)
 - ~0% wealth/property/inheritance tax
- Flexi-curity labor market (Not in this presentation)
 - high probability of being fired, but generous unemployment benefits
 - willing to adopt new technology: AI, Robots, China shock, Apple shock

Literature

- Equality and efficiency, appropriate institutions/policies
 - Marimon and Zilibotti (1999)
 - Song, Storesletten, Wang and Zilibotti (2015)
- Understanding different institutions and economies
 - Song, Storesletten, and Zilibotti (2011): Growing Like China
 - Fan, Peters and Zilibotti (2021): Growing Like India
- Can't We All Be More Like Scandinavians?
 - Yes, we can!

More Literature

- Education, Parenting and Altruism
 - Becker and Tomes (1979, 1986)
 - Doepke and Zilibotti (2017, 2019)
 - Song, Storesletten and Zilibotti (2012)
- Innovation
 - Acemoglu and Zilibotti (1997)
 - Acemoglu, Akcigit, Alp, Bloom, Kerr (2018)

Roadmap

- Introduction
- Benchmark model
- Extensions
- Conclusion

Benchmark Model: Environment

- Becker and Tomes (1979, 1986)
- Time: $t = 0, 1, 2, \dots$
- Two period OLG: childhood, adulthood
 - Adulthood: economically active
 - Childhood: only take education
- Three occupations L, H, E : Low-skilled, High-skilled, and entrepreneurs
- Two levels of education $e \in \{l, h\}$
- Each parent has one child, intergenerational correlation of ability

$$\ln z' = \rho \ln z + 1 - \rho + \varepsilon, \quad \varepsilon \sim N(0, \sigma_\varepsilon^2).$$

Ability matters for entrepreneurial skills and educational cost.

Benchmark Model: Adult

An adult with ability z , education e , and a child with ability z' , chooses occupation and child's education

- Occupation L if $e = l$
 - low skilled worker L : earns $(1 - \tau_L) w_L$
- Occupation H or E if $e = h$
 - high skilled worker H : earns $(1 - \tau_H) w_H$
 - entrepreneur E : with prob p , successfully builds a firm and gains $(1 - \tau_\pi) \pi(z)$;
with prob $1 - p$, fails and earns $(1 - \tau_L) w_L$

Benchmark Model: Adult

An adult, given z, z', e , chooses her child's education e' ,

- the parent cares about the child:

- child's income $\mathbb{E}y'(z', e')$, expected

- child's psychological cost of education $\kappa(z') = \lambda/z'$:

$$\theta \left[-\mathbf{1}_{e'=h} \kappa(z') + \mathbb{E}y'(z', e') \right]$$

- not child's utility derived from her altruism from grandchildren

- “Helicopter parenting”, Doepke and Zilibotti (2019)

- *not* tiger mom, *not* Song, Storesletten and Zilibotti (2015)

- Cost: financial cost χ

- the parent pays $\eta\chi$, and government pays $(1 - \eta)\chi$

Benchmark Model: Adult

- A low-education adult only chooses child's education

$$\begin{aligned} V_l(z, z') &= \max_{e'} u(c_L) + \theta [-\mathbf{1}_{e'=h} \kappa(z') + y'(z', e')] \\ \text{s.t. } c_L + \mathbf{1}_{e'=h} \eta \chi &= (1 - \tau_L) w_L. \end{aligned}$$

- A h-educated adult chooses occupation and child's education

$$\begin{aligned} V_h(z, z') &= \max \{ V_H(z, z'), V_E(z, z') \}. \\ V_H(z, z') &= \max_{e'} -\kappa(z) + u(c_H) + \theta [-\mathbf{1}_{e'=h} \kappa(z') + y'(z', e')] \\ \text{s.t. } c_H + \mathbf{1}_{e'=h} \eta \chi &= (1 - \tau_H) w_H. \\ V_E(z, z') &= \max_{e'} -\kappa(z) + p u(c_E) + (1 - p) u(c_L) \\ &\quad + \theta [-\mathbf{1}_{e'=h} \kappa(z') + y'(z', e')] \\ \text{s.t. } c_E + \mathbf{1}_{e'=h} \eta \chi &= (1 - \tau_\pi) \pi(z) \end{aligned}$$

Benchmark Model: Production

- An entrepreneur with ability z , with prob p , successfully builds a firm:

$$\pi(z) = \max_{n_H, n_L} z n_H^\alpha n_L^\beta - w_H n_H - w_L n_L$$

- FOCs imply

$$w_H n_H = \alpha z n_H^{\alpha-1} n_L^\beta,$$

$$w_L n_L = \beta z n_H^\alpha n_L^{\beta-1},$$

$$\pi(z) = (1 - \alpha - \beta) z n_H^\alpha n_L^\beta$$

$$= (1 - \alpha - \beta) z^{\frac{1}{1-\alpha-\beta}} \left(\frac{\beta}{w_L} \right)^{\frac{\beta}{1-\alpha-\beta}} \left(\frac{\alpha}{w_H} \right)^{\frac{\alpha}{1-\alpha-\beta}}.$$

Simple Example: Linear Utility

- Education decision only depends on z' but not z
- Simplified problem for h-educated parent:

$$V_H(z, z') = \max_{e' \in \{h, l\}} -\kappa(z) + \mathbb{E}c_h + \theta [-\mathbf{1}_{e'=H}\kappa(z') + y'(z', e')]$$

$$\text{s.t. } \mathbb{E}c_h + \mathbf{1}_{e'=h}\eta\chi = \max\{(1 - \tau_H)w_H, p(1 - \tau_\pi)\pi(z) + (1 - p)(1 - \tau_L)w_L\}$$

- Cutoff $\hat{z}$ for occupation decision

$$o = \begin{cases} E & \text{if } z \geq \hat{z} \\ H & \text{if } z < \hat{z} \end{cases}$$

where $\hat{z}$ is pinned down by

$$p(1 - \tau_\pi)\pi(\hat{z}) + (1 - p)(1 - \tau_L)w_L = (1 - \tau_H)w_H$$

Linear Utility: Education Decision

Cutoff $\bar{z}$ for education decision, same for h- and l-educated parents:

$$e = \begin{cases} h & \text{if } z' \geq \bar{z} \\ l & \text{if } z' < \bar{z} \end{cases}$$

where $\bar{z}$ is pinned down by

$$\frac{\lambda}{\bar{z}} = (1 - \tau_H)w'_H - (1 - \tau_L)w'_L - \frac{\eta\chi}{\theta}.$$

Linear Utility: Steady State

- The steady state exists if $\hat{z} > \bar{z}$ (guaranteed by choosing a not too high η)
- Market clearing conditions:

- High skilled labor supply = demand

$$F(\hat{z}) - F(\bar{z}) = p \int_{\hat{z}}^{\infty} z^{\frac{1}{1-\alpha-\beta}} \left(\frac{\alpha w_L}{\beta w_H} \right)^{\frac{1-\beta}{1-\alpha-\beta}} \left(\frac{\beta}{w_L} \right)^{\frac{1}{1-\alpha-\beta}} dF(z)$$

- Low skilled labor supply = demand

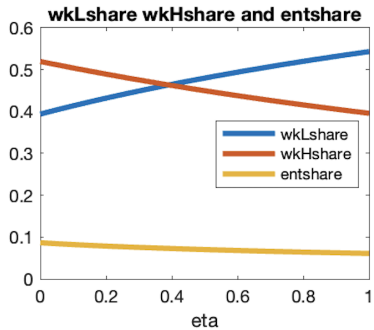
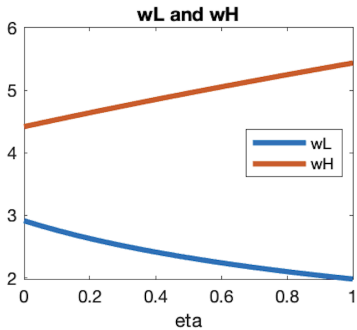
$$F(\bar{z}) + (1-p)(1-F(\hat{z})) = p \int_{\hat{z}}^{\infty} z^{\frac{1}{1-\alpha-\beta}} \left(\frac{\alpha w_L}{\beta w_H} \right)^{\frac{\alpha}{1-\alpha-\beta}} \left(\frac{\beta}{w_L} \right)^{\frac{1}{1-\alpha-\beta}} dF(z)$$

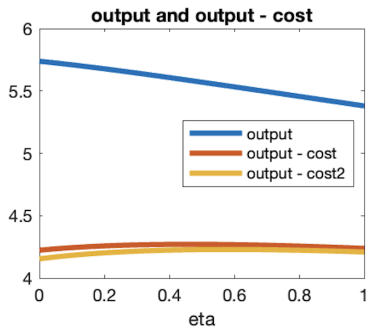
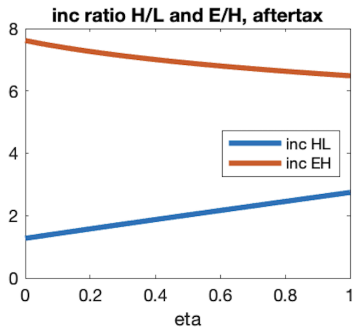
Linear Utility: Steady State Equilibrium

- Government budget

$$\begin{aligned}(1 - \eta)\chi(1 - F(\bar{z})) &= \tau_L w_L (F(\bar{z}) + (1 - \rho)(1 - F(\hat{z}))) \\ &\quad + \tau_H w_H (F(\hat{z}) - F(\bar{z})) + \tau_\pi p \int_{\hat{z}}^{\infty} \pi(z) dF(z) \\ &= \tau_L w_L (F(\bar{z}) + (1 - \rho)(1 - F(\hat{z}))) \\ &\quad + \gamma \tau_L w_H (F(\hat{z}) - F(\bar{z})) + \delta \tau_L p \int_{\hat{z}}^{\infty} \pi(z) dF(z)\end{aligned}$$

- Equilibrium: prices $\{w_H, w_L, \tau_L\}$ that clear the three constraints above
 - $\bar{z}, \hat{z}$ are endogenously determined
 - tax progressivity δ, γ are exogenously given here





Entrepreneurs in Laissez-Faire

$$\eta = 0, \tau = 0 \text{ (US?)}$$

- efficient allocation:

$$p\pi(z) + (1-p)w_L \geq w_H$$

- with risk (income and expenditure):

$$p u(c_E) + (1-p) u(c_L) \geq u(c_H),$$

$$c_E = \pi(z) - \mathbf{1}_{e'=h} \eta \chi,$$

$$c_L = w_L - \mathbf{1}_{e'=h} \eta \chi$$

Entrepreneurs with Safety Nets

- Safety net for entrepreneurs (Nordic)
 - $\eta = 0$, no/low expenditure risk
 - high w_L : low income risk if fails
- Additional benefits for entrepreneurs
 - low $w_H \Rightarrow$ high $\pi = y - w_H n_H - w_L n_L$
 - low τ_π

Education in Laissez-Faire

- Efficient allocation of education:

$$\frac{\lambda}{z'} \leq w'_H - w'_L - \chi$$

- Linear utility, Laissez-Faire ($\eta = 1, \tau = 0$):

$$\frac{\lambda}{z'} \leq w'_H - w'_L - \frac{\eta}{\theta} \chi,$$

- inefficiently low if $\theta < \eta$ (US?)
 - education determined by altruism but not social benefits/costs
- Concave utility, additional inefficiency

$$u(w_L - \eta \chi) + \theta \left[-\frac{\lambda}{\bar{z}} + w'_H \right] = u(w_L) + \theta w_L \Rightarrow$$
$$\frac{\lambda}{\bar{z}} = w'_H - w'_L - \frac{u(w_L) - u(w_L - \eta \chi)}{\theta}$$

Education: Free

- $\eta = 0$, $\tau_H > \tau_L$ (Nordic)
- Linear utility

$$\frac{\lambda}{\bar{z}} = (1 - \tau_H)w'_H - (1 - \tau_L)w'_L - 0 \cdot \frac{\chi}{\theta}$$

- Concave utility:

$$u((1 - \tau_L)w_L - 0 \cdot \chi) + \theta \left[-\frac{\lambda}{\bar{z}} + (1 - \tau_H)w_H \right] \geq u((1 - \tau_L)w_L) + \theta [(1 - \tau_L)w_L],$$
$$\frac{\lambda}{\bar{z}} = (1 - \tau_H)w'_H - (1 - \tau_L)w'_L - 0$$

- Can be efficient

Dynamics

- Start from an initial distribution of z_0, e_0
- Guess a sequence of $\{w_{Ht}, w_{Lt}, \tau_{Lt}\}_{t=0}^T$
- At time t , given the distribution of $\{z_t, e_t\}$
 - solve for occupation and education choices for each household type $\{z, e\}$, given prices and taxes at t and $t+1$
 - obtain e_{t+1} endogenously
 - compute z_{t+1} exogenously
- Continue until $t = T$
- Check labor market clearing conditions and government budget constraints from t to T .

Extensions and Future Work

- Alternative model setup closer to innovation literature
 - entrepreneurs and high-skilled workers produce high-tech product
 - low skilled workers produce low-tech products, same final goods
- Safety net in the labor market
 - low cost for workers to leave sunset industries/occupations
 - firms can flexibly adopt new technologies
 - “Pareto improvement” instead of “winners vs losers”
 - general support for Robots, AI, imports

Conclusion

- Nordic models
 - equality and inequality both exist
 - reduce risks for entrepreneurs
 - reduce costs of new technology
 - supported by appropriate education and labor market policies
- Implications for other countries during the new era of AI and imports